Short Abstract:

Over the past two decades, biomaterials research has advanced from foundational concepts to translatable approaches for repairing and regenerating tissues ranging from simple structures to complex organs. This progress has been driven by innovations in material design, microstructural control, bioinspired fabrication, and a deeper understanding of how physical and chemical cues regulate cell behavior and tissue formation. In our work, we have focused on developing novel biomaterials and engineered structures for the regeneration of large and challenging tissue defects. Our program includes: (i) large-area bone regeneration, using oxygen-tension-controlled matrices and bone–blood vessel cell synergy for vascularized bone formation; (ii) bone-cartilage interface formation, using complex structures with local microstructural control and tunable stiffness; (iii) biomimetic 3D printing, through decellularized matrices that better recapitulate native extracellular matrix; and (iv) strategies to promote repair and regeneration under disease conditions. Together, these efforts illustrate how advances in biomaterials design have expanded the possibilities for regenerative medicine. In this presentation, I will discuss the scientific and engineering principles underlying these studies and reflect on their contributions to the broader progress of the field.